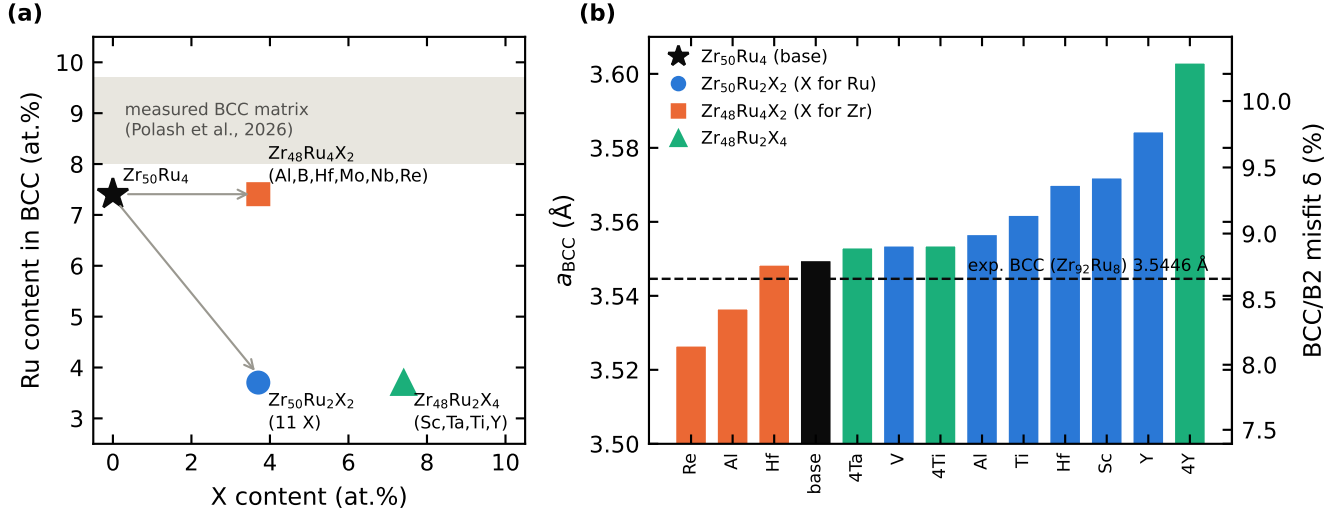
(a)
measured BCC matrix
(Polash et al., 2026)
Zr50Ru4
Zr48Ru4X2
(Al,B,Hf,Mo,Nb,Re)
Zr50Ru2X2
(11 X)
Zr48Ru2X4
(Sc,Ta,Ti,Y)
Ru content in BCC (at.%)
X content (at.%)
(b)
Zr50Ru4 (base)
Zr50Ru2X2 (X for Ru)
Zr48Ru4X2 (X for Zr)
Zr48Ru2X4
exp. BCC (Zr92Ru8) 3.5446 Å
aBCC (Å)
BCC/B2 misfit δ (%)
Re
Al
Hf
base
4Ta
V
4Ti
Al
Ti
Hf
Sc
Y
4Y